

SHING TAK LAM

Mathematics Institute, University of Warwick
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RESEARCH INTERESTS

Canonical metrics in Kähler geometry, Gauge theory, Geometric flows, Moment maps, Stability conditions and Moduli.

EDUCATION

University of Warwick 2025–28
Doctor of Philosophy (in progress)
PhD in Complex Differential Geometry supervised by Ruadhaí Dervan.
University of Glasgow 2024–25
First year of PhD in Complex Differential Geometry supervised by Ruadhaí Dervan.
Aligned student on the Algebra, Geometry and Quantum Fields CDT.
University of Cambridge 2023–24
Master of Mathematics *Distinction - 90%, ranked 17 out of 294*
Part III essay titled “Quantum cohomology and the Seidel representation” supervised by Jack Smith.
University of Cambridge 2020–23
Bachelor of Arts *First class honours*

PREPRINTS

Semistability and asymptotics of geometric flows, arXiv:2606.09235
Canonical metrics on families of vector bundles, Submitted arXiv:2512.04017

CONFERENCE TALKS

Fibrations and Deformations, Brest Jun 2025
Canonical Metrics on Families of Vector Bundles

JUNIOR SEMINAR TALKS

Junior Analysis and Probability Seminar, Warwick Jun 2026
Semistability and asymptotics of geometric flows
Hodge Club, Edinburgh Apr 2026
Groups acting on symplectic manifolds and moduli theory
Postgraduate Seminar, Warwick Nov 2025
Symmetries and dynamics - Introduction to symplectic geometry
Junior Algebraic Geometry Seminar, Warwick May 2025
Canonical metrics on families of vector bundles

OTHER TALKS

Reading group on K -stability of Fano varieties Jun 2026
 Θ -stratifications
Reading group on Gauge theoretic invariants Nov 2025
Compactness and wall-crossing
Reading group on Moduli of weighted hyperplane arrangements Nov 2025
Introduction to toric varieties
Reading group on Gauge theoretic invariants Oct 2025
Introduction to Gauge Theory and Seiberg–Witten invariants

Reading group on <i>K</i>-stability of Fano Varieties <i>The α- and δ-invariants</i>	<i>Jun 2025</i>
Glasgow Geometry and Topology Pre-Seminar <i>Gauge theory and Floer theory</i>	<i>Mar 2025</i>
Reading group on <i>Positivity in Algebraic Geometry</i> <i>Big line bundles, volume and Zariski decomposition</i>	<i>Feb 2025</i>
Reading group on <i>Algebraic Stacks</i> <i>Deformation theory and smoothness</i>	<i>Feb 2025</i>
Reading group on <i>Geometric Invariant Theory</i> <i>The Kempf-Ness theorem</i>	<i>Dec 2024</i>
AGQ CDT Examples showcase <i>Blowing up and rational elliptic surfaces</i>	<i>Nov 2024</i>
Part III Seminar <i>Symplectic reduction</i>	<i>Mar 2024</i>
Part III Seminar <i>Morse theory</i>	<i>Dec 2023</i>
Cambridge Summer Research Festival <i>Hyperkähler structures on nilpotent $SL(n, \mathbb{C})$ orbits</i>	<i>Oct 2023</i>
Cambridge Summer Research Festival <i>On generalised hardness of approximation in optimisation and data science</i>	<i>Oct 2022</i>

ORGANISATION

Reading group on <i>Geometric Invariant Theory</i>	<i>Autumn 2024</i>
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TEACHING

MA453 Lie Algebras Support Classes	<i>Term 2 2025-26</i>
MA259 Multivariable Calculus Marking	<i>Term 2 2025-26</i>

EXPERIENCE

Summer research internship , University of Cambridge Summer project supervised by Alexei Kovalev, studying hyperkähler metrics on coadjoint orbits.	<i>Summer 2023</i>
Summer research internship , University of Cambridge Summer project supervised by Anders Hansen, studying (non-)computability of optimisation problems.	<i>Summer 2022</i>

PRIZES AND SCHOLARSHIPS

Clemmow/B Talbot Prize, K Roth Prize <i>Awarded for highest first in Part II Mathematics at Peterhouse</i>	<i>2023</i>
Senior Burkhill Prize for Mathematics, Routh prize for Mathematics <i>Awarded for highest first in Part IB Mathematics at Peterhouse</i>	<i>2022</i>
Summer Research Bursary <i>Funding from the Faculty of Mathematics for summer research projects.</i>	<i>2022, 2023</i>
Scholarship in Mathematics <i>Scholar of Peterhouse, University of Cambridge</i>	<i>2021-24</i>
College prize <i>College prize for academic achievement</i>	<i>2021, 2022, 2023, 2024</i>

CONFERENCES ATTENDED

Degenerations in Complex Geometry , Institut Mittag-Leffler	<i>Jun-Jul 2026</i>
CALF , Nottingham	<i>Jun 2026</i>
Complex Geometry in Bordeaux , Bordeaux	<i>May 2026</i>

Living on the edge of the moduli space , Aarhus	<i>May 2026</i>
CALF , Imperial College London	<i>Mar 2026</i>
Brussels-London Geometry Seminar , Brussels	<i>Jan 2026</i>
Integrable Day 2025 , Loughborough	<i>Nov 2025</i>
COW , Warwick	<i>Nov 2025</i>
VII BrAG Meeting , Nottingham	<i>Sep 2025</i>
Singular canonical Kähler metrics on compact and non-compact manifolds , Budapest	<i>Sep 2025</i>
Summer Research Institute in Algebraic Geometry , Colorado	<i>Jul 2025</i>
Fibrations and Deformations , Brest	<i>Jun 2025</i>
Winter School on K-stability , CIRM, Luminy	<i>Mar 2025</i>
UKAG Network Winter School , Lancaster	<i>Dec 2024</i>
GLEN , Glasgow	<i>Dec 2024</i>

SKILLS

Programming Languages

Haskell, Python, Julia, Mathematica, Rust

Languages

Native: English, Cantonese, Mandarin

Basic: French, Spanish

Last updated: June 9, 2026